Henery Garção

Oceanographer · Environmental Modelling & Metocean Specialist

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PROFILE

Oceanographer and numerical modeller with 15 years of experience in hydrodynamic, wave, dispersion and flood modelling for coastal and offshore projects, from oil spill studies for environmental licensing in Brazil to metocean criteria in the Middle East. PhD candidate working on probabilistic flood risk.

EXPERIENCE

Modelling Coordinator · Umi San — Hidrografia e Engenharia, Brazil 2025 – present

- Leads hydrodynamic, wave and dispersion modelling for offshore and coastal projects.
- Coordinates environmental data analysis workflows, including metocean data visualisation tools.
- Contributes to R&D on smart meteoceanographic buoys for real-time coastal monitoring (Brazilian Symposium of Marine Geology and Geophysics, Vitória, 2025).

Senior Oceanographer · Fugro Middle East & India, Dubai, UAE 2023 – 2024

- Led metocean design criteria studies for offshore and coastal engineering projects in the Middle East and India.
- Developed regional SWAN wave hindcast models for the Arabian Gulf and Red Sea; automated modelling and data-processing workflows in Python and Matlab.
- Advised international clients on modelling studies; supported technical proposals and tenders.

Project Coordinator — Environmental Modelling · CLS Brasil (formerly Prooceano), Brazil 2015 – 2022

- Coordinated 40+ hydrodynamic, oil spill and dispersion modelling reports for offshore environmental licensing; managed teams of up to 8 and the interface with clients and regulators; presented results at public hearings.
- Developed the Fortran core of a search-and-rescue particle-tracking model, extended into the SPAD-SAR R&D programme (Shell · Prooceano · Brazilian Navy, ANP PD&I clause), as project coordinator and programmer.
- Technical coordinator of Projeto Costa Norte (2019–2022; Enauta · ANP · UFRJ · UFPA · UERJ); co-author of the Journal of Coastal Research article (2020) and of Chapter 8 (oil dispersion modelling) of the final report.
- Technical team member on Projeto Azul (Shell Brasil · COPPE/UFRJ · Prooceano, 2013–2022); co-author of the Elsevier book chapter (2015).
- Delivered training on oil spill dispersion modelling methodology (IBAMA licensing team, 2018) and on the OSCAR model (CLS headquarters, Toulouse, 2016).

Oceanographer · Prooceano, Brazil 2010 – 2015

- Hydrodynamic and Lagrangian oil dispersion modelling and data analysis for environmental licensing.
- Piloted Seaglider autonomous underwater vehicles for the Projeto Azul observation system.

EDUCATION

PhD, Environmental Engineering — UFES, 2025 – present (supervisor: Prof. José Antônio Tosta dos Reis; FAPES scholarship). Probabilistic framework for flood risk assessment in data-scarce watersheds, integrating hydrological and hydraulic modelling with stochastic input sampling and uncertainty propagation, yielding probabilistic flood inundation maps for risk management applications.

MSc, Environmental Engineering — UFES, 2008 – 2010 (ANP scholarship). Development of a Lagrangian particle model incorporating wind effects and oil slick spreading.

BSc, Oceanography — UFES, 2004 – 2007 (CNPq scholarship). Hydrodynamics and salinity distribution in Vitória Bay through computational modelling (2DH numerical model).

SELECTED PUBLICATIONS

De Carvalho, G.V.; Cabral, M.M.; De Oliveira, A.L.S.C.; **Garção, H.F.**; Pellegrini, J.A.C. Methodology to Evaluate the Coastal Susceptibility to Oil Spills Originated in Large Marine Areas — Costa Norte Project. *Journal of Coastal Research*, 95(SI):1344–1348, 2020. doi:10.2112/SI95-258.1

Santos, F.A.; De Oliveira, A.L.S.C.; … **Garção, H.F.**; … Fragoso, M.R. Projeto Azul: Operational Oceanography in an Active Oil and Gas Area, Southeastern Brazil. In: *Coastal Ocean Observing Systems*, Elsevier, 2015, pp. 99–116. doi:10.1016/B978-0-12-802022-7.00020-1

Oliveira, A.L.S.C.; Carvalho, G.V.; **Garção, H.F.**; Pellegrini, J.A.C.; Cabral, M.M. Modelagem de Dispersão de Óleo — Capítulo 8. In: *Projeto Costa Norte — Desenvolvimento de Metodologias para o entendimento de processos costeiros e estuarinos e da vulnerabilidade de florestas de mangue na Margem Equatorial Brasileira*, v.2. Enauta/ANP, 2020.

SKILLS

Models: Delft3D, SWAN, MEMW (OSCAR, DREAM/ParTrack), HEC-HMS / HEC-RAS, CROCO (recently started), MOHID (course)

Programming and tools: Python, Fortran, Matlab, Git/GitHub, QGIS, ArcGIS, Surfer

Applications: metocean criteria, extreme value analysis, oil spill and pollutant dispersion, Lagrangian transport, coastal and estuarine hydrodynamics, probabilistic flood modelling

TRAINING

Nature-based Solutions for Disaster and Climate Resilience — SDG Academy, online, 2024

Metocean Criteria and Fugro's Software — Fugro, Wallingford (training trip), 40 h, 2023

Coastal Systems Modelling with MOHID — University of São Paulo, 120 h, 2008

MEMW Training Course — SINTEF, Trondheim, 32 h, 2011

Fundamentals of Oil Spill Response; Advanced Oil Spill Modeling and Data Sources — International Oil Spill Conference, 2 × 2 h, 2014

Delivered: Oil Spill Dispersion Modelling Methodology, IBAMA licensing team, 2018 · OSCAR model, CLS headquarters, Toulouse, 24 h, 2016

LANGUAGES

Portuguese (native) · English (proficient) · Spanish (basic)